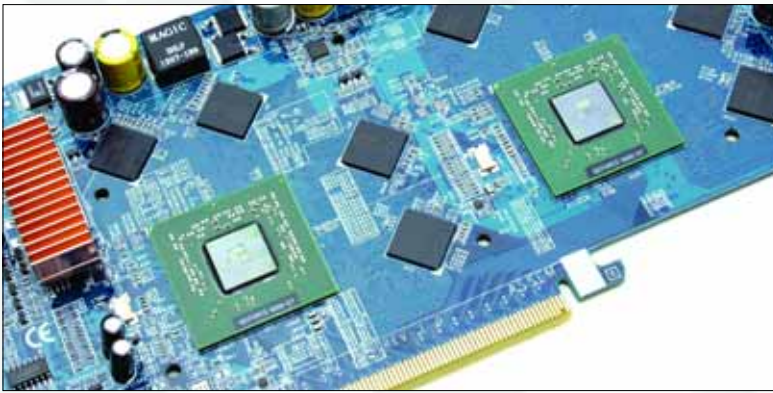




Here's a better picture of dual-GPUs on a single card, with the cooling fans removed

Basically, dual-GPU cards work exactly the same way as two cards in SLI or CrossFire mode, except that the card's memory is shared, and transfer between the chips is much faster, due to the fact that they're on the same PCB.

Dual-Core GPUs?

One would think that, logically, the advent of dual-core CPUs should now prompt graphics card manufacturers to shift towards dual-core GPUs as well. This doesn't seem to be the case, though.

nVidia, for one, has made it clear on several occasions that they currently have no interest in dual-core GPUs, simply because they are not needed. According to tests and benchmarks, two nVidia cards in SLI mode perform more floating point operations than the best Intel and AMD

dual-core CPUs! So as far as current technologies go, it would be an understatement to say that GPU technology is way ahead of the CPU market!

Conclusion

Unfortunately, neither Intel nor AMD can match dual-GPU speeds, and essentially, the graphics sub-system is bottlenecked by the system architecture. This happens because it is the system CPU that send the requests for graphics processing to the GPUs, and thus far, even the best CPUs cannot do this fast enough. Perhaps dual-core CPUs will minimise the loss of speed we have faced.

Of course, by the time dual-core CPUs become the norm, the champions of graphics (nVidia and ATi) will have fought each other into a new era of GPUs all over again—if history is anything to go by. Overall, it's a great fight that's going on between these giants and it's us, the end-users, who will benefit. So far, a multi-GPU system—whether two cards or one—is the way to go. If you're the type who needs the best visual feel possible, you need to get yourself one!

It looks like parallel processing for graphics is here to stay, whether in the form of two cards running in parallel, or two chips on a single card parallel-processing textures and frames. We're not naïve enough to make this a long-term prediction, simply because the graphics market is one of the most competitive, and technologies come and go almost on a monthly basis.

So we'll leave you with a blanket statement: "If there's a change in this trend, we'll be the first to let you know." ☒

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